

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks:30

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and

authentication methods. Explain how your solution balances coverage, performance, security, and cost.

3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

ANSWERS

Below is a rubric-oriented answer for all three case studies. Each solution identifies the constraints, decomposes the problem, proposes a feasible design, and justifies the choices.

1. Rural Branch Office – Internet Connectivity and Security

Problem Analysis

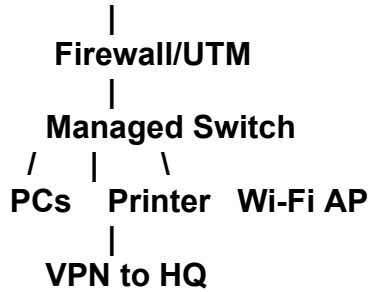
The branch office has:

- No high-speed wired Internet.
- Need for cloud applications and video conferencing.
- Need for secure communication with headquarters.
- Limited connectivity budget.
- Requirement for uninterrupted business operations.
- Need for future expansion.

Recommended Network Design

Satellite / 4G/5G Internet

|
Dual-WAN Router



Recommended Components

Component	Recommendation	Reason
Primary Internet	4G/5G fixed wireless	Lower deployment cost than wired infrastructure where cellular coverage is available
Backup Internet	Satellite or second cellular provider	Provides redundancy
Router	Dual-WAN business router	Automatically switches between connections
Firewall	Next-generation firewall/UTM	Protects the branch from Internet threats
Switch	Managed Gigabit switch	VLANs and network management
Wi-Fi	Business-grade Wi-Fi 6 AP	Supports employees and mobile devices
HQ connection	Site-to-site VPN	Encrypts branch-to-HQ traffic
UPS	UPS for networking equipment	Maintains operation during short power interruptions

Bandwidth Strategy

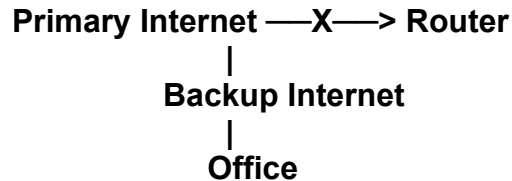
Because bandwidth may be limited, the company should prioritize business traffic:

1. Cloud applications and VPN traffic get high priority.
2. Video conferencing gets sufficient QoS priority.
3. Non-business downloads and streaming are restricted.
4. Usage monitoring helps identify bandwidth problems.

Reliability

A dual-WAN configuration is recommended. The primary connection can be 4G/5G, while a secondary provider or satellite connection acts as backup.

If the primary connection fails:



Traffic can automatically move to the backup connection.

Security

- Site-to-site IPsec VPN between branch and headquarters.
- Next-generation firewall.
- WPA3 Enterprise for wireless access.
- Strong administrator authentication.
- Regular firmware and security updates.
- Separate guest Wi-Fi from company devices.
- Endpoint protection on employee computers.

Scalability

The managed switch should have additional ports, and the router/firewall should support increased bandwidth and additional VPN connections.

Justification

A 4G/5G primary connection with a backup connection, dual-WAN router, firewall, and VPN provides a reasonable balance between cost, reliability, security, and performance. It avoids waiting for expensive wired infrastructure while allowing the branch to expand later.

2. University Wireless Network – 3,000+ Students

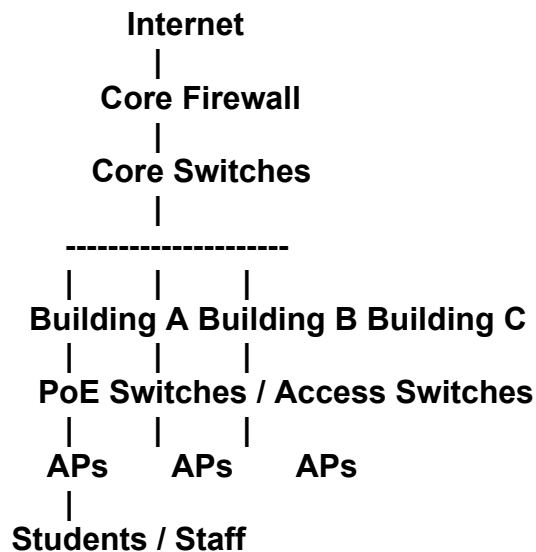
Problem Analysis

The university requires:

- More than 3,000 simultaneous users.
- Coverage across classrooms, laboratories, libraries, and hostels.
- Limited number of access points.
- Low interference.
- Secure authentication.
- Good performance at reasonable cost.

The key challenge is that coverage alone is not enough. High-density areas need sufficient wireless capacity.

Proposed Architecture



Access Point Placement

APs should be placed after a wireless site survey, rather than simply placing one AP per building or floor.

Priority should be given to:

- Classrooms with high student density.
- Lecture halls.
- Libraries.
- Laboratories.
- Hostel common areas.
- Corridors and outdoor student areas where required.

High-density locations should receive more AP capacity than low-usage areas.

Frequency Selection

Band

Use

2.4 GHz **Longer range and compatibility with older devices**

5 GHz **Main band for high-performance student access**

6 GHz **Use where supported by Wi-Fi 6E/7 clients and infrastructure**

The design should avoid excessive use of 2.4 GHz because it has fewer non-overlapping channels and is more susceptible to interference.

Channel Planning

Instead of allowing every AP to use the same channel:

AP1 → Channel A

AP2 → Channel B

AP3 → Channel C

AP4 → Channel A

AP5 → Channel B

A centrally managed wireless controller can automatically optimize channels and transmit power.

Authentication and Security

For students and staff:

- **WPA2/WPA3-Enterprise**
- **802.1X authentication**
- **RADIUS authentication server**
- **Individual student/staff credentials**

For guests:

- **Separate guest SSID.**
- **Internet-only access.**
- **Client isolation where appropriate.**

VLAN Design

Student Wi-Fi → Student VLAN

Staff Wi-Fi → Staff VLAN

Guest Wi-Fi → Guest VLAN

IoT Devices → IoT VLAN

This prevents unnecessary communication between different user groups.

Balancing Coverage, Performance, Security and Cost

Requirement	Solution
Coverage	Strategic AP placement
High density	Additional APs in crowded areas
Interference	Channel and power management
Performance	Prefer 5 GHz/6 GHz
Security	WPA3/802.1X/RADIUS
Cost	APs concentrated where demand is highest
Scalability	Central wireless management

Justification

With a limited AP budget, the university should not attempt to maximize physical coverage at the expense of capacity. APs should be concentrated in high-density locations, with centrally managed channel allocation and transmit power. This provides better performance for 3,000+ users while avoiding unnecessary AP deployment.

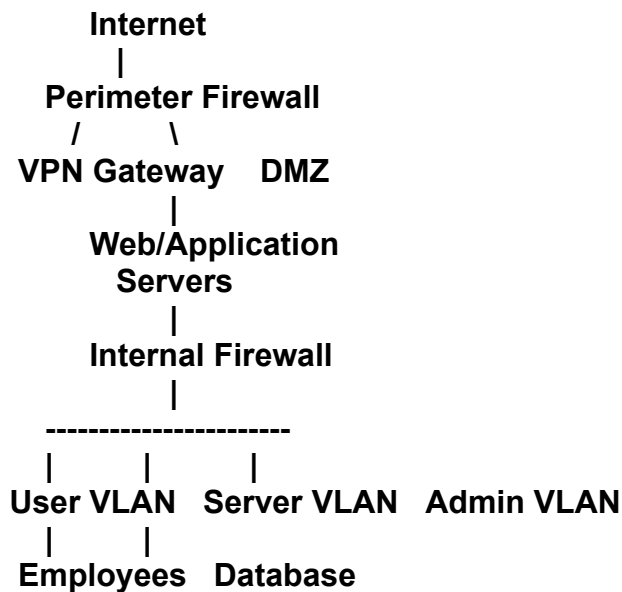
3. Financial Institution – Secure Network Architecture

Problem Analysis

The financial institution requires:

- Strong cybersecurity.
- Protection of sensitive customer information.
- Access to online banking services.
- Strict security and compliance requirements.
- High network performance.
- Controlled operational costs.

Proposed Architecture



Network Segmentation

The network should be divided into separate security zones.

Zone	Purpose	
User VLAN	Employee computers	
Server VLAN	Internal servers	application
Database VLAN	Sensitive databases	customer
Management VLAN	Network administration	
Guest VLAN	Visitor Internet access	
DMZ	Public-facing services	

The database VLAN should have the strongest restrictions.

For example:

Employee → Application Server → Database
✓ ✓ ✓

Employee → Database directly

x

This reduces the possibility of unauthorized access.

Firewall Deployment

Use multiple firewall/security layers:

- 1. Perimeter firewall protects the organization from Internet traffic.**
- 2. Internal segmentation firewall controls traffic between sensitive network zones.**
- 3. Web Application Firewall (WAF) protects public-facing web applications where required.**

Firewall rules should follow the least-privilege principle: only necessary traffic is allowed.

Authentication

For employees accessing sensitive systems:

- Multi-factor authentication (MFA).**
- Strong password policies.**
- Role-based access control.**
- 802.1X for network access.**
- Privileged access management for administrators.**

Example:

Employee



Username + Password



MFA Verification



Role Verification



Authorized Application

Intrusion Detection and Prevention

Deploy IDS/IPS to monitor suspicious traffic.

The system can identify:

- Port scanning.**
- Malware traffic.**
- Unauthorized access attempts.**

- Suspicious network behavior.
- Known attack signatures.

Security logs should be centralized for monitoring and investigation.

Performance Considerations

Security controls should be placed strategically rather than inspecting unnecessary traffic repeatedly.

- Use hardware-accelerated firewalls where appropriate.
- Apply QoS to critical applications.
- Use VLANs to reduce unnecessary broadcast traffic.
- Monitor firewall and IDS/IPS performance.
- Use redundant security devices for critical services.

Cost Control

Instead of deploying expensive security appliances everywhere:

- Concentrate advanced security controls around critical systems.
- Use VLANs and access-control policies for logical segmentation.
- Centralize logging and monitoring.
- Automate security alerts and routine administration.
- Use existing managed switches where they meet security requirements.

Compliance

The institution should map its security controls to the specific regulations and standards applicable to its jurisdiction and banking operations. Security policies should include:

- Access control.
- Data protection.
- Audit logging.
- Incident response.
- Regular vulnerability assessment.
- Backup and recovery.
- Periodic security audits.

Justification

The proposed architecture uses segmentation, layered firewalls, MFA, role-based access, and IDS/IPS to protect sensitive financial information. At the same time, VLANs, centralized management, and selective inspection help control operational costs and avoid unnecessary performance overhead.

Overall Comparison

Requirement	Rural Branch	University	Financial Institution
Main challenge	Limited Internet availability	High-density wireless users	Security of sensitive data
Primary solution	4G/5G + backup	Managed high-density Wi-Fi	Segmented secure architecture
Reliability	Dual-WAN	Redundant network infrastructure	Redundant security devices
Security	Firewall + VPN	WPA3/802.1X	MFA + firewalls + IDS/IPS
Scalability	Additional WAN/LAN capacity	Add APs based on demand	Additional VLANs/security zones
Cost strategy	Wireless instead of expensive wired deployment	APs concentrated by demand	Layered security with centralized management

Final Conclusion

The three scenarios require different network strategies because their constraints are different.

- The rural branch should prioritize affordable connectivity and redundancy using 4G/5G, backup connectivity, dual-WAN, firewall, and VPN.
- The university should prioritize wireless capacity using strategic AP placement, 5/6 GHz, channel management, and WPA2/WPA3-Enterprise with 802.1X.
- The financial institution should prioritize security through network segmentation, layered firewalls, MFA, RBAC, and IDS/IPS.

Thus, the best network design is not simply the most expensive one; it is the design that satisfies the required performance, reliability, security, scalability, and budget constraints.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

